

Visualization Building Blocks & Scenarios

A Card Game-Based Learning Framework for the InfoVis Design Space

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Use the three ingredients of the framework to create game variants for exercises that fit your lecture and challenge your students as they progress

Step 1: Describe the learning goals, expected cognitive skills and topics

Remember

Understand

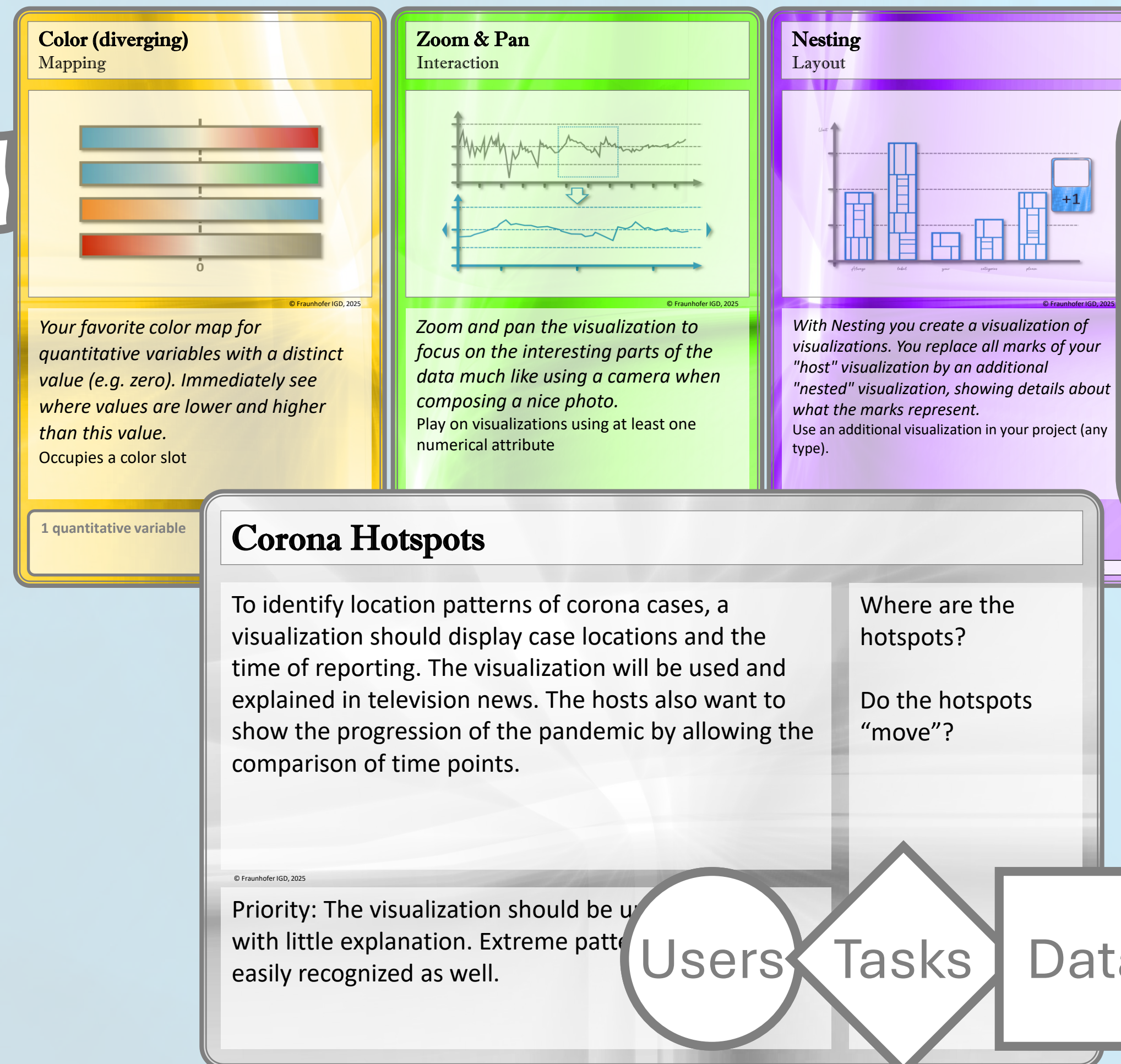
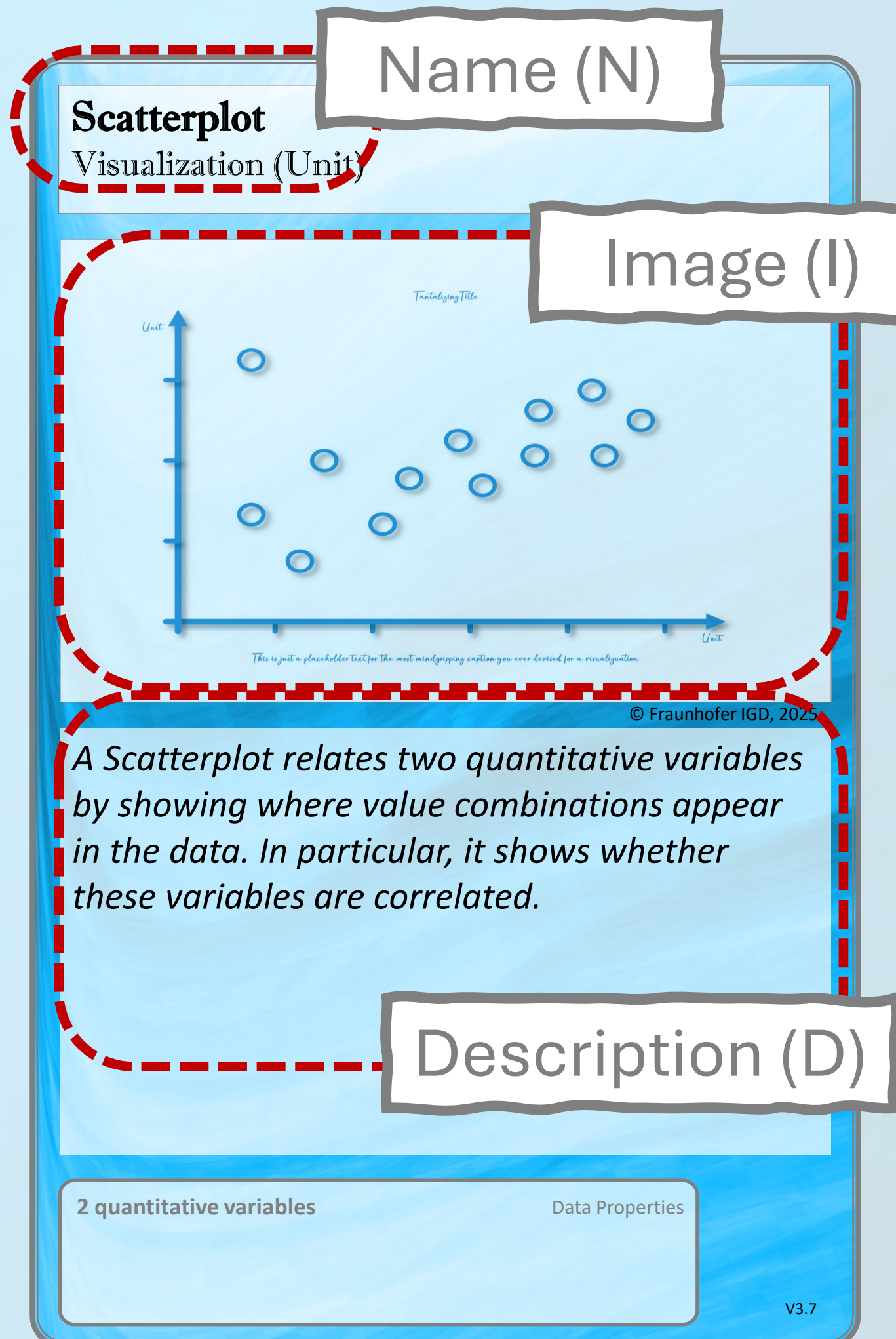
Apply

Analyze

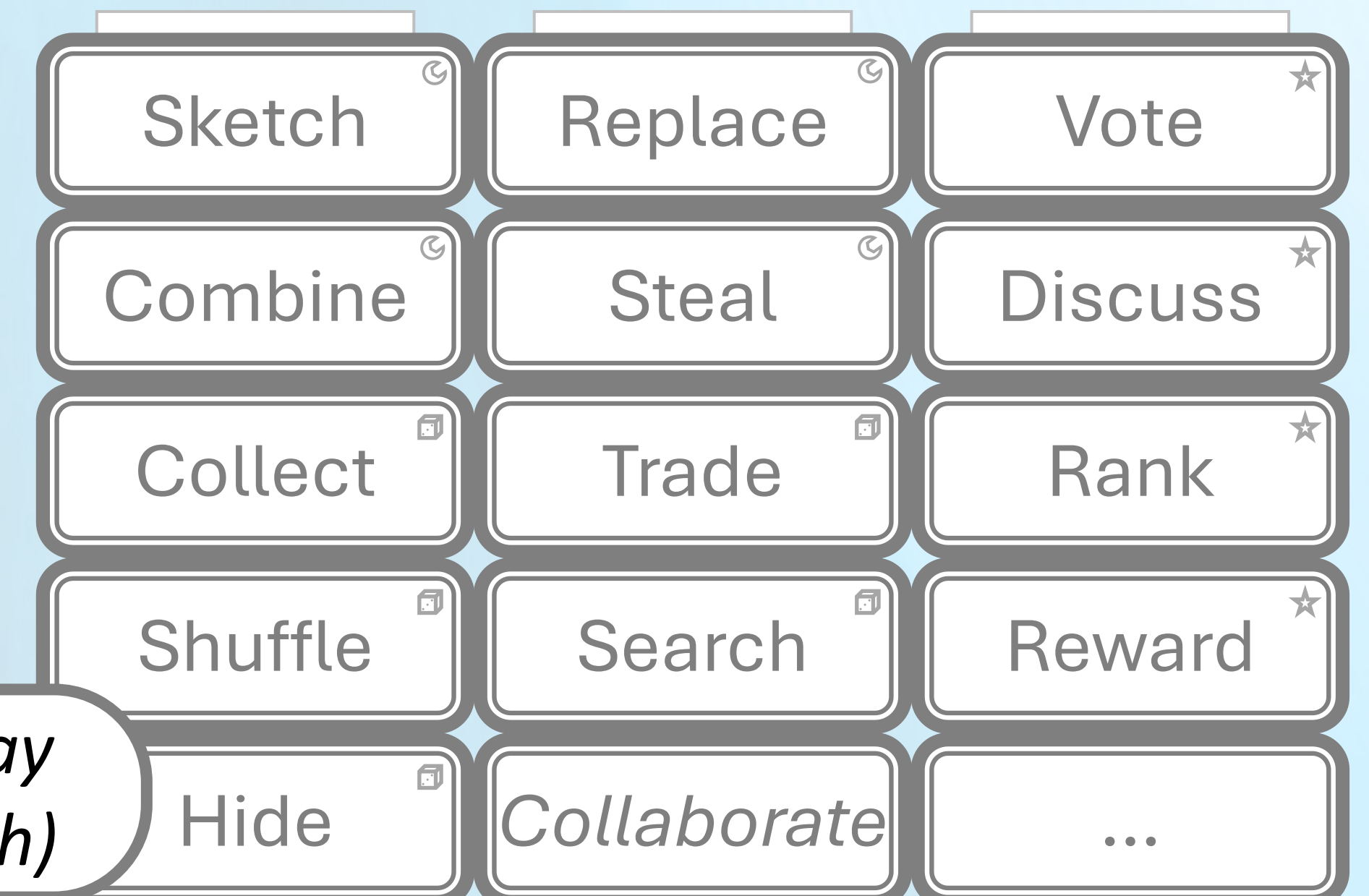
Evaluate

Create

cf. Bloom's revised taxonomy

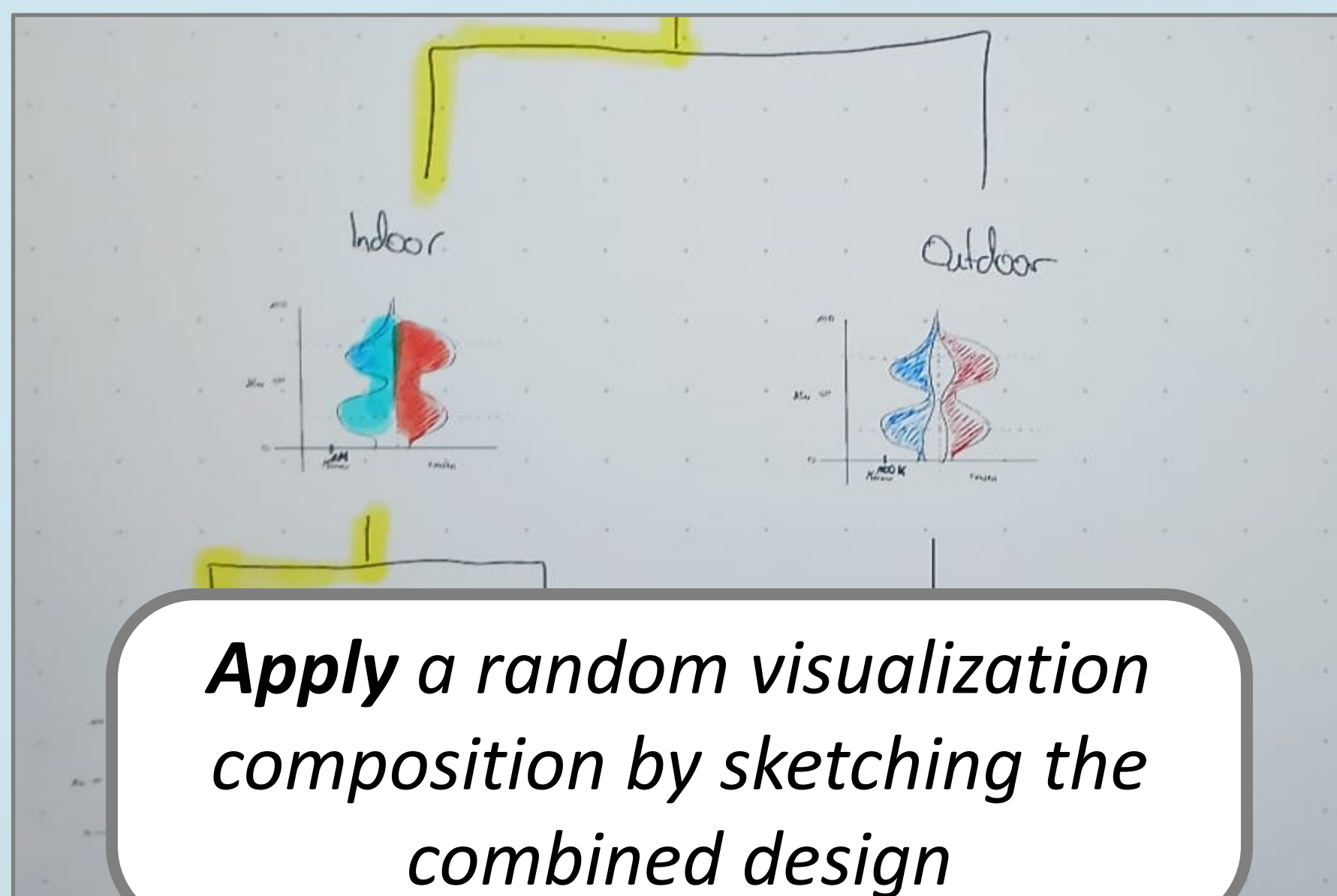


Step 2: Choose from **visualization**, **mapping**, **interaction** and **layout** cards (70+ in total). Use **scenarios** to challenge learners with (sometimes) conflicting requirements on users, tasks and data.



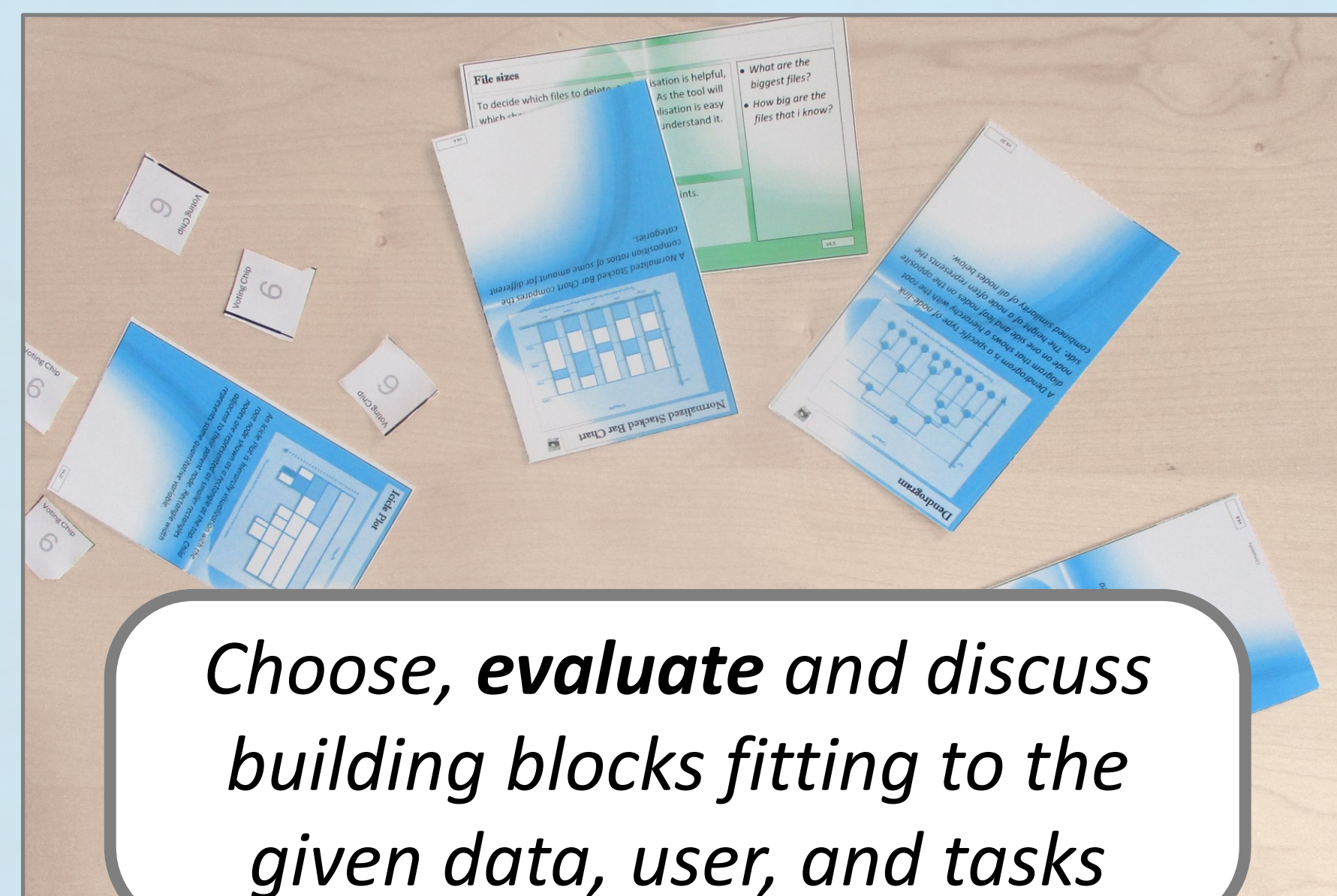
Step 3: Create the exercise using game mechanics to hone the skills and incentives to play „winning“ designs or explanations. (Understanding this step is at the core of our research)

Example 1: VisCompose



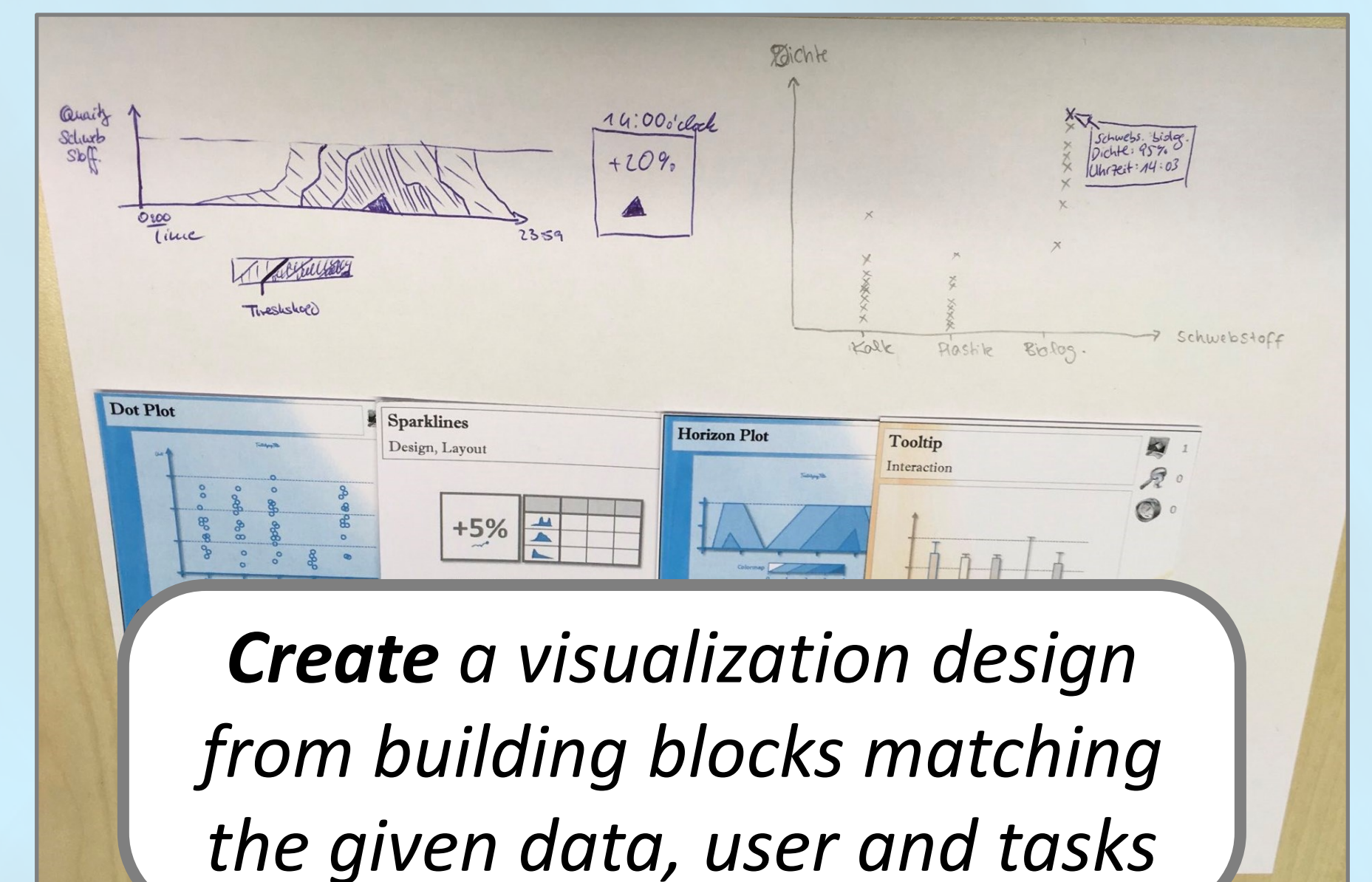
Apply a random visualization composition by sketching the combined design

Example 2: VisCussion



Choose, **evaluate** and discuss building blocks fitting to the given data, user, and tasks

Example 3: ViSketch



Create a visualization design from building blocks matching the given data, user and tasks

Students Experiencing the Design Space

- Building blocks and scenarios for open learning games
- Combine building blocks to tackle scenarios and assess their appropriateness to user, task and data requirements
- Experience the design space through imagining or creating designs from self-chosen, given, or random building blocks
- Tailor games towards a specific scope and learning goals to fit different stages of a curriculum

Used in five courses with over 180 BSc., MSc. and PhD Students over two years

Current Research: How to Fit a Game to Learning Goals

- Understand how different games may reflect specific learning goals, topics and levels of experience throughout an entire course
- Understand the limits of how game mechanics embed rules of „good‘ design

Observed Benefits

- Provides scaffolding for students against the problem of „no hard rules‘ of vis design
- Exercises keep students activated with minimal involvement of the lecturer

What our Students said...

- „Instead of using pure intuition or convention, you are forced to weigh pros and cons to argue in the group.“
- „I do not remember all names of techniques but the required verbalization [of appropriateness] helps when recalling the concepts.“
- „Discussions in the group help connecting concepts to experiences.“
- „To be able to sort through a bunch of different plot options, it is way easier to have those cards handy.“

References

- He and Adar; VizItCards: A Card-Based Toolkit for Infovis Design Education; *IEEE TVCG*, 23(1), 2017
- Keck et al.; A Didactic Framework for Analyzing Learning Activities to Design InfoVis Courses; *IEEE CGA*, 41(6), 2021